

Stage 2: Behavioral Feature Set

C22 Retail Fade Strategy

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1. Design Principles

The feature library operationalizes the Stage 1 EDA findings under three constraints fixed before implementation: (1) **strict causality** — every feature is computable at position open time from prior information only, enforced mechanically by a test that recomputes features on truncated trade sequences and asserts bitwise identity; (2) **streaming-native architecture** — features are produced by a `TraderState` class with a `compute-then-update` interface that processes one position at a time, so the identical code serves batch research and the Stage 4 real-time engine with no reimplementations; (3) **pre-registration** — the feature list was specified in a frozen document (`FEATURES.md`) before implementation; the only post-hoc modification is redundancy pruning under a pre-committed $|\rho| > 0.9$ rule (28 specified $\rightarrow$ 21 retained), with no additions. Submitted code comprises the feature library (`features.py`), evaluation harness (`splits.py`), build script, and the causality test suite; all validation tables in the appendices are reproduced by a single command.

2. Feature Catalogue

21 features in five families. Each cites its Stage 1 evidence.

- **A — Streak & momentum** (EDA §3, confirmed triggers): `loss_streak`, `win_streak`, `pnl_ewm` (recursive exponentially-weighted P&L, $\alpha = 0.3$).
- **B — Sizing behavior** (EDA §4, size–streak interaction): `lot_zscore` (size vs. trader’s own running distribution), `amount` (raw), `size_after_loss_delta` (escalation vs. pre-streak baseline).
- **C — Risk discipline** (EDA §3, stop-loss findings): `has_sl`, `has_tp`, `sl_distance_pct`, `sl_usage_rate_5`, `manual_exit_rate_5` (rolling 5-trade discipline trends).
- **D — Session & pressure** (EDA §1–2): `pnl_pct`, `dd_from_peak_pct`, `trade_index`, `log_dt_close`, `trades_per_hour`.
- **E — Identity & context** (EDA §2.1): `prior_campaigns` (strictly earlier campaigns only), `shared_ip`, `ip_cluster_size`, `challenge_type`, `gold_vol_prev_day` (prior calendar day’s realized range; same-day values excluded as lookahead).

Deliberate exclusions. `session_pct` and `trades_remaining_est` (require the session’s final extent — lookahead); `reverseProfit` and any transform of it (target leakage: it is an exact linear function of the target); same-day volatility (lookahead). Documenting exclusions is part of the audit trail: every candidate rejected for causality is listed with its reason in `FEATURES.md`.

3. Univariate Validation Highlights

Full tables in Appendix B; primary era (C53–C65, 6,582 positions). Selected results, expected gross loss per lot vs. the \$7.00 hurdle:

- **Stop distance is the strongest single discipline signal**, monotonic across all five quintiles: tightest stops $-\$63.0/\text{lot}$ (these traders profit gross) to widest stops $+\$113.0/\text{lot}$. This continuous measure subsumes the binary stop-loss flag: on the per-lot basis, the Stage 1 no-SL dollar edge is confirmed to be a tail phenomenon — equal-weighted, no-SL positions average only $+\$7.2/\text{lot}$ — while the discipline gradient migrates to *how* stops are placed rather than *whether* they are placed.
- **Streaks remain the core state variable**: `loss_streak > 1` yields $+\$40.5/\text{lot}$ (reproducing the Stage 1 confirmed trigger through independent reimplementations), while `win_streak > 1` yields $-\$22.6/\text{lot}$ — a symmetric *do-not-fade* condition. The Stage 3 score is therefore naturally two-sided: abstaining on winning streaks is a free improvement over an unconditional fade baseline.
- **Experience monotonically erodes the edge**: first-campaign participants average $+\$21.3/\text{lot}$; participants with more than three prior campaigns average $-\$11.6/\text{lot}$. The fade pool is disproportionately fresh participants — with the corollary that each trader’s data value to C22 decays with repeated participation.
- **The fade zone is mild frustration, not desperation**: positions opened at slightly negative session P&L (0 to -1.65%) average $+\$34/\text{lot}$, while deeply underwater positions average near zero — consistent with the Stage 1 finding that deep drawdown alone fails the hurdle.
- **Rapid opening bursts are a do-not-fade state**: 175 positions (142 accounts) were opened at instantaneous rates above 60 trades/hour — almost all second or third entries placed within minutes of the session’s first trade. These traders *win* ($-\$73.9/\text{lot}$): quick multi-entry opens appear to reflect pre-planned executions of a single trade idea (consistent with the challenge’s 3-minute idea-grouping rule) rather than tilt. The `trades_per_hour` feature captures this as an early-session burst signal; sustained high-frequency activity deeper into sessions does not appear in the data.

4. Walk-Forward Evaluation Harness

Stage 3 evaluation uses expanding-window, campaign-blocked folds within C53–C65 (4 folds). Because 68.2% of active traders recur across campaigns, any `traderKey` present in a training fold is removed from that fold’s validation set. Decomposing the purge showed that *every* trader-overlap case also shares an IP cluster with its earlier appearance — repeat participants return on the same infrastructure — so blanket IP-level purging would only compound the same removals while also excluding unrelated users of shared networks. Consistent with the Stage 1 synchrony analysis, IP-level purging is therefore applied only to the seven cluster-campaign pairs that showed elevated trading synchrony. Post-purge validation sets contain 371–901 positions (43–61% row attrition), which we report plainly: the purged sets are thin but leakage-free, and Stage 3 will report both purged and unpurged metrics — the gap between them measures how much model performance depends on trader recurrence.

5. Redundancy & Retained Correlations

Seven specified features were pruned under the pre-committed rule (deterministic duplicates and $|\rho| > 0.9$ near-duplicates). Three moderate correlations are retained deliberately: `loss_streak`–`dd_from_peak_pct` ($\rho = 0.74$) and `pnl_ewm`–`pnl_pct` ($\rho = 0.85$) are related but distinct pressure measures; `lot_zscore`–`size_after_loss_delta` ($\rho = 0.86$) separate relative size from streak-conditional escalation, the axis the Stage 1 interaction finding identified. Stage 3's regularized model is the correct place to arbitrate these, not premature manual exclusion.

Appendix A — Feature Definitions

Retained Stage 2 behavioral features (21).

Feature	Family	Definition	Causality note / Stage 1 evidence
<code>loss_streak</code>	A	Consecutive positions with net P&L < 0 before this one	Prior outcomes only; memo §3, confirmed triggers 1–2
<code>win_streak</code>	A	Consecutive positions with net P&L > 0 before this one	Prior outcomes only; memo §3
<code>pnl_ewm</code>	A	EW sum of past net P&L, $\alpha = 0.3$: $e_t = 0.3 p_{t-1} + 0.7 e_{t-1}$	Prior realized P&L only; memo §3
<code>lot_zscore</code>	B	(amount – running mean)/running std of prior amounts; NaN if < 3 prior	Trader's own prior sizing history; memo §4
<code>amount</code>	B	Raw lot size (pass-through)	Observed at open; memo §4
<code>size_after_loss_delta</code>	B	Amount minus mean of pre-streak amounts; NaN if no streak or no pre-streak trades	Pre-open streak state and prior amounts only; memo §4
<code>has_sl</code>	C	Stop-loss set on opening ticket	Open-time field; memo §3
<code>has_tp</code>	C	Take-profit set on opening ticket	Open-time field; memo §3
<code>sl_distance_pct</code>	C	$ \text{openPrice} - \text{slPrice} /\text{openPrice}$; NaN if no SL	Open-time fields only; memo §3
<code>sl_usage_rate_5</code>	C	Fraction of last 5 prior positions with SL set; NaN if < 3 prior	Rolling prior-trade rate; memo §3
<code>manual_exit_rate_5</code>	C	Fraction of last 5 prior positions closed manually; NaN if < 3 prior	Rolling prior exits only; memo §3
<code>pnl_pct</code>	D	Cumulative net P&L / starting balance	As-of open, pre-outcome; memo §1–2
<code>dd_from_peak_pct</code>	D	(Peak equity – equity)/starting balance	As-of open, pre-outcome; memo §1–2
<code>trade_index</code>	D	1-based position count within account-campaign	Open-time count; memo §1–2
<code>log_dt_close</code>	D	$\ln(1 + \text{seconds since last close})$; NaN if first trade	Last realized close only; memo §1–2
<code>trades_per_hour</code>	D	<code>trade_index</code> / $\max(\text{hours since first open}, 1/60)$	Elapsed time only; memo §1–2
<code>prior_campaigns</code>	E	Distinct earlier campaigns in which this traderKey traded	Strictly earlier campaigns; memo §2, §2.1
<code>shared_ip</code>	E	Shared-IP flag (sanitized meta-data)	Static join; memo §2.1
<code>ip_cluster_size</code>	E	Active accounts sharing the anonymized IP cluster	Static infrastructure property; memo §2.1

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Feature	Family	Definition	Causality note / Stage 1 evidence
challenge_type	E	Challenge type id; 'unknown' if absent	Static join; memo §2
gold_vol_prev_day	E	(High–Low)/Close of XAUUSD, prior calendar day	Prior day only; same-day excluded as lookahead

Appendix B — Univariate Validation

Primary era (C53–C65): mean gross loss per lot by feature bucket. Buckets in ascending numeric order.

Feature	Bucket	<i>n</i>	Mean loss/lot
loss_streak	≤ 1	5701	8.09
	(1, 9]	881	40.54
win_streak	≤ 1	5779	17.30
	(1, 10]	803	–22.59
pnl_ewm	(–470.8, –31.4]	1317	–3.35
	(–31.4, 0]	3201	33.89
	(0, 21.7]	747	8.27
	(21.7, 363.3]	1317	–21.55
lot_zscore	(–30.55, –0.99]	476	–16.52
	(–0.99, –0.56]	475	11.01
	(–0.56, –0.04]	476	–9.73
	(–0.04, 0.97]	475	1.52
	(0.97, 53.0]	476	13.15
amount	(0.01, 0.15]	1402	49.77
	(0.15, 0.32]	1263	–2.89
	(0.32, 0.57]	1307	8.90
	(0.57, 0.97]	1301	–0.94
	(0.97, 6.5]	1309	4.06
size_after_loss_delta	(–2.59, –0.34]	264	–21.60
	(–0.34, –0.11]	264	11.21
	(–0.11, 0.01]	266	27.55
	(0.01, 0.28]	259	–3.47
	(0.28, 5.74]	263	24.53
has_sl	False	1569	7.17
	True	5013	14.08
has_tp	False	1314	11.20
	True	5268	12.74
sl_distance_pct	(0, 0.00028]	1003	–63.00
	(0.00028, 0.00059]	1002	–14.31
	(0.00059, 0.0010]	1003	11.49
	(0.0010, 0.0016]	1002	23.24
	(0.0016, 0.903]	1003	112.98
sl_usage_rate_5	(0, 0.6]	665	–5.77
	(0.6, 0.8]	529	–16.41
	(0.8, 1.0]	1211	11.52
manual_exit_rate_5	(0, 0.4]	574	13.39

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Feature	Bucket	<i>n</i>	Mean loss/lot
pnl_pct	(0.4, 0.6]	428	47.40
	(0.6, 0.8]	720	−13.92
	(0.8, 1.0]	683	−24.19
	(−0.215, −0.017]	1317	0.50
	(−0.017, 0]	3202	34.33
	(0, 0.0099]	746	−6.53
dd_from_peak_pct	(0.0099, 0.138]	1317	−18.12
	(0, 0.004]	3949	14.86
	(0.004, 0.023]	1316	13.18
	(0.023, 0.267]	1317	4.44
trade_index	(1, 2]	3351	25.46
	(2, 3]	826	−5.91
	(3, 6]	1245	−21.96
	(6, 57]	1160	24.81
log_dt_close	(0, 5.17]	2679	−4.97
	(5.17, 7.60]	893	−10.25
	(7.60, 10.93]	893	16.91
trades_per_hour	(0.09, 1.21]	1317	15.43
	(1.21, 4.38]	1316	−7.65
	(4.38, 26.57]	1316	−8.04
	(26.57, 60.0]	2458	38.70
	(60.0, 161.2]	175	−73.87
	≤ 1	4030	21.31
prior_campaigns	(1, 3]	1356	7.28
	(3, 16]	1196	−11.62
shared_ip	False	953	13.60
	True	5629	12.24
ip_cluster_size	(1, 3]	2797	22.40
	(3, 7]	1274	3.14
	(7, 24]	1224	−20.75
	(24, 64]	1272	33.21
challenge_type	11	2589	10.85
	unknown	3993	13.47
gold_vol_prev_day	(0.006, 0.018]	1432	−0.13
	(0.018, 0.020]	1658	0.72
	(0.020, 0.021]	1048	28.02
	(0.021, 0.033]	1554	22.12
	(0.033, 0.050]	890	19.21

Appendix C — Walk-Forward Fold Composition

Primary-era expanding-window folds; validation purged of traderKeys present in training (plus seven synchrony-flagged IP-cluster exceptions).

Fold	Train campaigns	Val cam- paings	Val rows	Rows kept	Attrition	Val accts	Accts kept
1	53–56	57–58	949	371	60.9%	234	123
2	53–58	59–60	1033	454	56.1%	296	166
3	53–60	61–62	973	464	52.3%	280	146

Fold	Train campaigns	Val paings	cam-	Val rows	Rows kept	Attrition	Val accts	Accts kept
4	53–62	63–65		1590	901	43.3%	360	249